



Barker College

Student Number

**2007
TRIAL
HIGHER SCHOOL
CERTIFICATE**

Mathematics

Staff Involved:

AM TUESDAY 31 JULY

- JML* • LMD
- AJD* • MRB
- RMH • JM
- GDH • EAS
- PJR • JGD

155 copies

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Write your Barker Student Number on all pages of your answers
- Board-approved calculators may be used
- Start each question on a NEW page
- A table of standard integrals is provided at the back of this paper which may be detached for your use
- All necessary working must be shown in every question
- Marks may be deducted for careless or badly arranged working
- Write on one side of the page only

Total marks – 120

- Attempt Questions 1 – 10
- All questions are of equal value

Total marks – 120
Attempt Questions 1–10
All questions are of equal value

Answer each question on a new page.

Marks

Question 1 (12 marks)

- | | | |
|-----|--|---|
| (a) | Evaluate $\sqrt{\frac{3 \cdot 8^3 \times 93}{3518}}$ correct to two decimal places. | 2 |
| (b) | Make 'a' the subject of the formula $s = ut + \frac{1}{2}at^2$ | 2 |
| (c) | Show on a number line the solution of $ 2m - 1 \leq 5$ | 2 |
| (d) | If $G(x) = 1 - x^2$, evaluate $G(3) - G(0)$ | 2 |
| (e) | Factorise completely $3x^2 - 12y^2$ | 2 |
| (f) | The probability of gaining relief from a particular illness with antibiotic A is $\frac{7}{10}$,
whilst with antibiotic B, the probability of gaining relief is $\frac{3}{4}$.
One cold sufferer takes A, whilst another takes B.
What is the probability that only one cold sufferer gains relief? | 2 |

Question 2 (12 marks)		Marks
[START A NEW PAGE]		
(a)	Find the equation of the tangent to the curve $y = x^2 - 3x + 2$ at the point (3, 2)	2
(b)	Differentiate with respect to x :	
(i)	$\tan(\log_e x)$	2
(ii)	$x\sqrt{1+x}$	2
(c)	(i) Find a primitive function of $\frac{2x}{x^2 + 1}$	1
	(ii) Evaluate $\int_{\frac{\pi}{3}}^{\pi} \cos\left(\frac{1}{2}x\right) dx$	2
(d)	Given that $f(x) = x^2 + x$, find the values of b for which $f''(b) = f(b)$	3

Question 3 (12 marks)		Marks
[START A NEW PAGE]		
(a)	Evaluate $\log_2 48 - \log_2 4 - \log_2 3$	2
(b)	Two perpendicular lines $3x + 2y = 12$ and $2x + ay = b$ intersect at the point (2, 3). Find the values of a and b .	2
(c)	The sum of the first three terms of an arithmetic sequence is 27. The sum of the next three terms is 63. Find the first term and common difference.	3
(d)	Evaluate $\sum_{r=1}^{20} (2r + 1)$	2
(e)	By first writing the recurring decimal $0.303030\dots$ as an infinite geometric series, use your knowledge of geometric series to express it as a rational number in its simplest form.	3

Question 4 (12 marks)

[START A NEW PAGE]

Marks

The coordinates of the points A, B and C are (0, 2), (4, 0) and (6, -4) respectively.

(i) Find the length AB.

2

(ii) Find the gradient of AB.

1

(iii) Show that the equation of the line ℓ , drawn through C and parallel to AB is

$$x + 2y + 2 = 0$$

2

(iv) Find the coordinates of D, the point where ℓ intersects the x -axis.

2

(v) Find the perpendicular distance of the point A from the line ℓ .

2

(vi) Hence, or otherwise, find the area of the quadrilateral ABCD.

3

Question 5 (12 marks)

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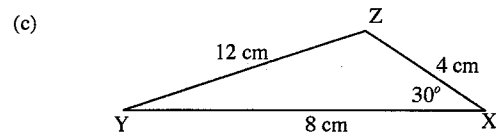
Marks

(a) Show that $(\sin A + \cos A)^2 + (\sin A - \cos A)^2 = 2$

2

(b) Solve $2\cos x = \sqrt{3}$, where $0 \leq x \leq 2\pi$

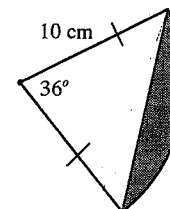
2



Find the area of the triangle XYZ.

2

(d)



The diagram shows the area of a minor segment of a circle with radius 10 cm.

Find the area of this minor segment.

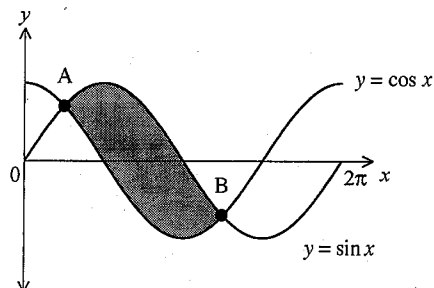
2

Question 5 continues on next page

Question 5 (continued)

Marks

(e)



The diagram shows the graphs of the curves $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq 2\pi$

- (i) Show that the x -coordinates of the two marked points A and B are

$\frac{\pi}{4}$ and $\frac{5\pi}{4}$ respectively.

1

- (ii) Calculate the shaded area leaving your answer in surd form.

3

End of Question 5

Question 6 (12 marks)

[START A NEW PAGE]

Marks

- (a) Find all values of k for which the quadratic equation $kx^2 - 8x + k = 0$ has real roots.

3

- (b) A function $f(x)$ is defined by $f(x) = 2x^3 + ax^2 + bx + 3$

- (i) If this function $y = f(x)$ has stationary points when $x = 1$ and $x = -2$, show that $a = 3$ and $b = -12$.

2

- (ii) Hence, find the coordinates of the stationary points of $y = f(x)$ and determine their nature.

3

- (iii) Hence, find the coordinates of the point of inflexion.

2

- (iv) Hence, sketch the graph $y = f(x)$ identifying all key features, i.e. showing the turning points and the point of inflexion.

1

- (v) For what values of x is $f(x) = x^3 + 3x^2 - 12x + 3$ concave up?

1

(a)

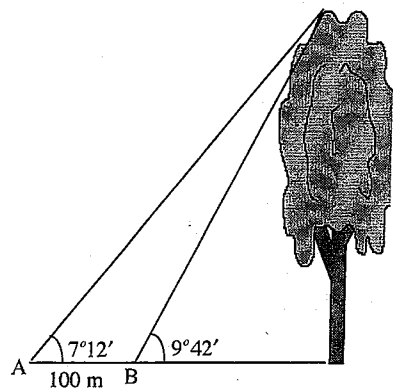


DIAGRAM NOT DRAWN
TO SCALE

The diagram above was sketched by a surveyor, who measured the angle of elevation of the top of a tree on the other side of a river to be $7^\circ 12'$ from point A.

From point B, 100 metres directly towards the tree from A, the angle of elevation of the top of the same tree was $9^\circ 42'$.

- (i) Show that the height of the tree can be expressed in the form 3

$$h = \frac{100 \sin 7^\circ 12' \sin 9^\circ 42'}{\sin 2^\circ 30'}$$

- (ii) Calculate the height of the tree correct to three significant figures. 1

- (b) (i) Find the x -intercepts of the curve $y = 2x - x^2$ 1

- (ii) The area bounded by the curve $y = 2x - x^2$ and the x -axis is rotated about the x -axis. 3

Find the volume of the solid of revolution formed, leaving your answer in terms of π .

Question 7 continues on next page

- (c) Consider the function given by $f(x) = \sin^2 x$.

- (i) Copy and complete the following table onto **your answer page**. 2

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π
$y = f(x)$					

- (ii) Apply Simpson's rule with five function values to find an approximation 2

to $\int_0^\pi \sin^2 x \, dx$, leaving answer in exact form.

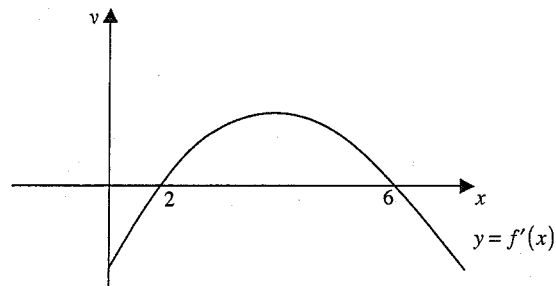
End of Question 7

Question 8 (12 marks)

[START A NEW PAGE]

Marks

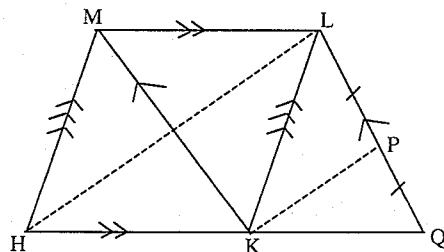
(a)



The diagram shows the graph of the gradient function of the curve $y = f(x)$.

- (i) For what values of x does $y = f(x)$ have a local minimum? Justify your answer. 2
- (ii) Draw a possible sketch of the curve $y = f(x)$. 1

(b)



HKLM is a parallelogram.

The line through L parallel to MK meets HK produced at Q.

- (i) Prove that $HK = KQ$. 2
- (ii) Given that P is the mid-point of LQ, prove that $\angle PKQ = \angle LHQ$. 3

- (c) On a table there are two bags A and B. Bag A contains three red and five blue marbles, whilst bag B contains one red and two blue marbles.

First Ann draws one marble from bag A at random and puts it in bag B.

Then Ann draws two marbles from bag B at random without replacement.

- (i) What is the probability that a marble drawn from bag A is blue? 1
- (ii) Using a tree diagram, or otherwise, find the probability that both marbles drawn from bag B are blue? 3

Question 9 (12 marks)

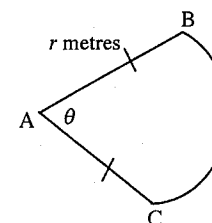
[START A NEW PAGE]

Marks

- (a) A horticulturist found the probability that a planted tomato seedling will eventually bear fruit was 0.75. He planted 'n' seedlings.

- (i) What is the probability that no seedlings will bear fruit? 1
- (ii) How many seedlings must be planted to be at least 99% certain that at least one seedling will bear fruit? 3

(b)



In the figure, AB and AC are radii of length r metres, of a circle with centre A.

The arc BC of the circle subtends an angle of θ radians at the centre A.

- (i) Write down the formulae for:
 - (α) the length of the arc BC 1
 - (β) the area of the sector ABC 1
- (ii) The perimeter of the sector ABC is 12 metres.
 - (α) Show that the area Y of the sector is given by $Y = \frac{72\theta}{(\theta+2)^2}$ 2
 - (β) Hence, show that the maximum area of the sector is 9 square metres. 4

- (a) (i) A mathematically-minded ant of negligible size sets out on a walking adventure on a number plane. 4

Starting at the origin, it walks out a distance of one unit in the positive direction along the x -axis.

It then turns left 90° and goes up half a unit from its current point.

If the ant continues turning left 90° , going half of the distance it previously went and keeps repeating this pattern, determine the coordinates where the ant will eventually end up.

- (ii) Explain why these coordinates can only be considered as close approximations. 1

- (b) Robert takes out a \$250 000 mortgage and agrees to pay the bank a \$2000 installment each month. The interest rate on the loan is 7.2% per annum, compounded monthly, and the contract requires that the loan is paid off within 20 years.

It can be shown that the amount A_n owing after the n^{th} repayment is given by the formula:

$$A_n = P(1+r)^n - \frac{M((1+r)^n - 1)}{r}$$

where P is amount borrowed

r is the interest rate per compounding period

M is the amount of each installment.

n is the number of compounding periods

- (i) Find the amount owing on the loan at the end of the 10th year. 1

- (ii) Find A_{240} and explain why this shows that the loan is actually paid off in less than 20 years. 2

- (iii) Find how many months early that the loan is paid off. 4
(ie. how many months below 20 years)

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, \quad x > 0$

2007 TRIAL HSC 2U MATHEMATICS SOLN'S

QUESTION 1

$$= 1.204395022$$

$$= 1.20 \text{ (2dp)}$$

$$s = ut + \frac{1}{2}at^2$$

$$s - ut = \frac{1}{2}at^2$$

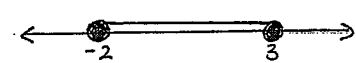
$$2(s - ut) = at^2$$

$$\frac{2(s - ut)}{t^2} = a$$

$$-5 \leq 2m - 1 \leq 5$$

$$-4 \leq 2m \leq 6$$

$$-2 \leq m \leq 3$$



$$G(x) = 1 - x^2$$

$$G(3) = 1 - 3^2 = -8$$

$$G(0) = 1 - 0^2 = 1$$

$$G(3) - G(0) = -8 - 1$$

$$= -9$$

$$= 3(x^2 - 4y^2)$$

$$= 3(x - 2y)(x + 2y)$$

$$= P(R_A N_B) + P(N_A R_B)$$

$$= \frac{7}{10} \times \frac{1}{4} + \frac{3}{10} \times \frac{3}{4}$$

$$= \frac{2}{5}$$

QUESTION 2

$$y = x^2 - 3x + 2$$

$$\frac{dy}{dx} = 2x - 3$$

$$x = 3 \quad \frac{dy}{dx} = 2(3) - 3$$

$$= 3 = m$$

$$\text{then } m = 3 \quad \text{pt } (3, 2)$$

$$y - 2 = 3(x - 3)$$

$$y - 2 = 3x - 9$$

$$y = 3x - 7 \quad \text{or } 3x - y - 7 = 0$$

$$(b)(i) y = \tan(\ln x)$$

$$\frac{dy}{dx} = \frac{1}{x} \sec^2(\ln x)$$

$$(ii) y = x(1+x)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = (1+x)^{\frac{1}{2}} \cdot 1 + x \cdot \frac{1}{2}(1+x)^{-\frac{1}{2}}$$

$$= \sqrt{1+x} + \frac{x}{2\sqrt{1+x}}$$

$$(c)(i) \int \frac{2x}{x^2+1} dx = \ln(x^2+1) + C$$

$$(ii) \int_{\frac{\pi}{3}}^{\pi} \cos\left(\frac{1}{2}x\right) dx$$

$$= 2 \left[\sin\left(\frac{1}{2}x\right) \right]_{\frac{\pi}{3}}^{\pi}$$

$$= 2 \left[\sin \frac{\pi}{2} - \sin \frac{\pi}{6} \right]$$

$$= 2 \left[1 - \frac{1}{2} \right]$$

$$= 1$$

$$(d) f(x) = x^2 + x$$

$$f'(x) = 2x$$

$$f''(x) = 2$$

$$f(b) = b^2 + b$$

$$f''(b) = 2$$

$$2 = b^2 + b$$

$$0 = b^2 + b - 2$$

$$0 = (b+2)(b-1)$$

$$\therefore b = -2, 1$$

QUESTION 3

$$(a) \log_2 48 - \log_2 4 - \log_2 3$$

$$= \log_2 48 - (\log_2 4 + \log_2 3)$$

$$= \log_2 \left(\frac{48}{4 \times 3} \right)$$

$$= \log_2 4$$

$$= 2 \log_2 2$$

$$= 2$$

$$2y = -3x + 12 \quad ay = -2x + b$$

$$y = -\frac{3}{2}x + 6 \quad y = -\frac{2}{a}x + \frac{b}{a}$$

$$m_1 = -\frac{3}{2} \quad m_2 = -\frac{2}{a}$$

$$h \Rightarrow m_1 \times m_2 = -1$$

$$-\frac{3}{2} \times -\frac{2}{a} = -1$$

$$\frac{6}{2a} = -1$$

$$6 = -2a$$

$$a = -3$$

$$\text{Sub } (2, 3) \quad 2(2) + -3(3) = b$$

$$4 - 9 = b$$

$$b = -5$$

$$a = -3, b = -5$$

$$(c) a + a + d + a + 2d = 27$$

$$3a + 3d = 27$$

$$a + d = 9 \Rightarrow a = 9 - d$$

$$a + 3d + a + 4d + a + 5d = 63$$

$$3a + 12d = 63$$

$$a + 4d = 21$$

$$(9 - d) + 4d = 21$$

$$9 + 3d = 21$$

$$3d = 12$$

$$d = 4 \quad a = 9 - 4$$

$$(d) \sum_{r=1}^{20} (2r+1) = 3 + 5 + 7 + \dots + 41$$

$$S_{20} = \frac{20}{2} (3 + 41)$$

$$= 10 \times 44$$

$$= 440$$

$$e) 0.\dot{3}0 = 0.30 + 0.0030 + 0.000030 + \dots$$

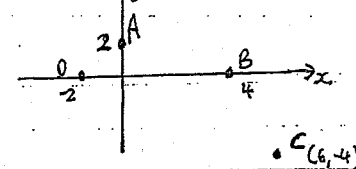
$$a = 0.30 \quad r = 0.01 \quad n = \infty$$

$$S_{\infty} = \frac{0.30}{1 - 0.01}$$

$$= \frac{30}{99}$$

$$= \frac{10}{33}$$

QUESTION 4



$$(i) d_{AB} = \sqrt{(4-0)^2 + (0-2)^2}$$

$$= \sqrt{20}$$

$$= 2\sqrt{5}$$

$$(ii) m_{AB} = \frac{0-2}{4-0}$$

$$= -\frac{1}{2}$$

$$(iii) l \Rightarrow m = -\frac{1}{2} \quad c(6, 4)$$

$$y - 4 = -\frac{1}{2}(x - 6)$$

$$2y + 8 = -x + 6$$

$$x + 2y + 2 = 0$$

$$(iv) l \Rightarrow x - mt \Rightarrow y = 0$$

$$x + 2(0) + 2 = 0$$

$$x = -2 \quad \therefore (-2, 0)$$

$$(v) d = \frac{|1(0) + 2(2) + 2|}{\sqrt{1^2 + 2^2}}$$

$$= \frac{6}{\sqrt{5}}$$

$$= \frac{6\sqrt{5}}{5}$$

$$= \frac{6}{\sqrt{5}}$$

$$(vi) ABCD \Rightarrow \text{Trapezium}$$

$$d_{DC} = \sqrt{(6-2)^2 + (-4-0)^2}$$

$$= \sqrt{80}$$

$$= 4\sqrt{5}$$

$$A = \frac{1}{2} \times \frac{6}{\sqrt{5}} \times [2\sqrt{5} + 4\sqrt{5}]$$

$$= \frac{3}{\sqrt{5}} \times 6\sqrt{5}$$

$$= 18u^2$$

QUESTION 5

$$(a) LHS = \sin^2 A + 2 \sin A \cos A + \cos^2 A$$

$$+ \sin^2 A - 2 \sin A \cos A + \cos^2 A$$

$$= 2 \sin^2 A + 2 \cos^2 A$$

$$= 2 (\sin^2 A + \cos^2 A)$$

$$= 2 \times 1 = 2 = RHS$$

$$2 \cos x = \sqrt{3}$$

$$\cos x = \frac{\sqrt{3}}{2}$$

$$x = \frac{\pi}{6}, \frac{11\pi}{6}$$

$$A = \frac{1}{2} \times 4 \times 8 \times \sin 30^\circ$$

$$= 8 \text{ cm}^2$$

$$36^\circ = \frac{36\pi}{180} = \frac{\pi}{5}$$

$$A = \frac{1}{2} \times 10^2 \left(\frac{\pi}{5} - \sin \frac{\pi}{5} \right)$$

$$= 10\pi - 50 \sin \frac{\pi}{5}$$

$$= 2.026663921$$

$$= 2 \text{ cm}^2 \text{ (n.cm)}$$

$$y = \sin x \quad y = \cos x$$

$$\sin x = \cos x$$

$$\tan x = 1$$

$$x = \frac{\pi}{4}, \frac{5\pi}{4}$$

$$(ii) A = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x) dx$$

$$= \left[-\cos x - \sin x \right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$= - \left[\cos x + \sin x \right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$= - \left[\left(\cos \frac{5\pi}{4} + \sin \frac{5\pi}{4} \right) - \left(\cos \frac{\pi}{4} + \sin \frac{\pi}{4} \right) \right]$$

$$= - \left[\left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) - \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) \right]$$

$$= - \left[-\frac{2}{\sqrt{2}} - \frac{2}{\sqrt{2}} \right]$$

$$= \frac{4}{\sqrt{2}} u^2$$

$$= \frac{4}{\sqrt{2}} u^2$$

$$= \frac{4}{\sqrt{2}} u^2$$

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$$= \frac{4}{\sqrt{2}} u^2$$

$$(b)(i) f(x) = 2x^3 + ax^2 + bx + 3$$

$$f'(x) = 6x^2 + 2ax + b$$

$$f'(-2) = 6(-2)^2 + 2a(-2) + b$$

$$= 24 - 4a + b$$

$$f'(1) = 6(1)^2 + 2a(1) + b$$

$$= 6 + 2a + b$$

$$24 - 4a + b = 0 \quad 6 + 2a + b = 0$$

$$-4a + b = -24 \quad 2a + b = -6$$

$$-4a + b = -24 \quad 2a + b = -6$$

$$-6a = -18$$

$$a = 3$$

$$2(3) + b = -6$$

$$6 + b = -6$$

$$b = -12$$

$$(ii) f(1) = 2(1)^3 + 3(1)^2 - 12(1) + 3$$

$$= -4$$

$$f(-2) = 2(-2)^3 + 3(-2)^2 - 12(-2) + 3$$

$$= 23$$

$$f''(x) = 12x + 6$$

$$f''(1) = 12(1) + 6$$

$$= 18 > 0 \quad \therefore \text{Min Pt}$$

$$f''(-2) = 12(-2) + 6$$

$$= -18 < 0 \quad \therefore \text{Max Pt}$$

$$(iii) f''(x) = 12x + 6 = 0$$

$$12x + 6 = 0$$

$$12x = -6$$

$$x = -\frac{1}{2}$$

$$f\left(-\frac{1}{2}\right) = 2\left(-\frac{1}{2}\right)^3 + 3\left(-\frac{1}{2}\right)^2 - 12\left(-\frac{1}{2}\right) + 3$$

$$= 9\frac{1}{2}$$

$$x \mid -1 \quad -\frac{1}{2} \quad 0$$

$$f'(x) \mid -12+6 \quad -6+6 \quad 0+6$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

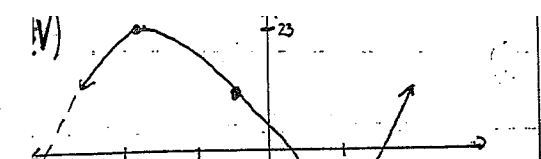
$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$

$$= -6 < 0 \quad = 0 \quad = 6 > 0$$



$$(v) x > -\frac{1}{2}$$

$$QUESTION 7$$

$$x \mid 0 \quad \frac{\pi}{4} \quad \frac{\pi}{2} \quad \frac{3\pi}{4} \quad \pi$$

$$y \mid 0 \quad \frac{1}{2} \quad 1 \quad \frac{1}{2} \quad 0$$

$$(ii) A = \frac{\pi}{3} \left[0 + 0 + 4\left(\frac{1}{2} + \frac{1}{2}\right) + 2(1) \right]$$

$$= \frac{\pi}{12} \times 6$$

$$= \frac{\pi}{2}$$

$$QUESTION 8$$

$$(a)(i) \text{ Stat Pts at } x=2 \text{ \& } x=6$$

$$x \mid 1 \quad 2 \quad 3 \quad x \mid 5 \quad 6 \quad 7$$

$$f(x) \mid - \quad 0 \quad + \quad f(x) \mid + \quad 0 \quad -$$

$$\backslash \quad - \quad / \quad \quad \quad / \quad - \quad \backslash$$

$$\text{Min at } x=2 \quad \text{Max at } x=6$$

$$(ii) h = 48 - 41268995$$

$$= 48.4 \text{ m (3sf)}$$

$$(b)(i) x\text{-Int} \Rightarrow y=0$$

$$0 = 2x - x^2$$

$$0 = x(2-x)$$

$$x=0 \quad x=2$$

$$(ii) V = \pi \int_0^2 (2x - x^2)^2 dx$$

$$= \pi \int_0^2 (4x^2 - 4x^3 + x^4) dx$$

$$= \pi \left[\frac{4x^3}{3} - x^4 + \frac{x^5}{5} \right]_0^2$$

$$= \pi \left[\left(\frac{32}{3} - 16 + \frac{32}{5} \right) - (0) \right]$$

$$= \pi \left(\frac{16}{15} \right)$$

$$= \frac{16\pi}{15} u^3$$

$$(c)(i) x \mid 0 \quad \frac{\pi}{4} \quad \frac{\pi}{2} \quad \frac{3\pi}{4} \quad \pi$$

$$y \mid 0 \quad \frac{1}{2} \quad 1 \quad \frac{1}{2} \quad 0$$

$$(ii) A = \frac{\pi}{3} \left[0 + 0 + 4\left(\frac{1}{2} + \frac{1}{2}\right) + 2(1) \right]$$

$$= \frac{\pi}{12} \times 6$$

$$= \frac{\pi}{2}$$

$$QUESTION 8$$

$$(a)(i) \text{ Stat Pts at } x=2 \text{ \& } x=6$$

$$x \mid 1 \quad 2 \quad 3 \quad x \mid 5 \quad 6 \quad 7$$

$$f(x) \mid - \quad 0 \quad + \quad f(x) \mid + \quad 0 \quad -$$

$$\backslash \quad - \quad / \quad \quad \quad / \quad - \quad \backslash$$

$$\text{Min at } x=2 \quad \text{Max at } x=6$$

$$(ii) h = 48 - 41268995$$

$$= 48.4 \text{ m (3sf)}$$

$$(b)(i) x\text{-Int} \Rightarrow y=0$$

$$0 = 2x - x^2$$

$$0 = x(2-x)$$

$$x=0 \quad x=2$$

$$(ii) V = \pi \int_0^2 (2x - x^2)^2 dx$$

$$= \pi \int_0^2 (4x^2 - 4x^3 + x^4) dx$$

$$= \pi \left[\frac{4x^3}{3} - x^4 + \frac{x^5}{5} \right]_0^2$$

$$= \pi \left[\left(\frac{32}{3} - 16 + \frac{32}{5} \right) - (0) \right]$$

$$= \pi \left(\frac{16}{15} \right)$$

$$= \frac{16\pi}{15} u^3$$

$$(b)(i) ML \parallel KQ \text{ (given)}$$

$$MK \parallel LQ \text{ (given)}$$

$$KQLM \text{ is } \parallel \text{ gram (Opp sides } \parallel \text{)}$$

$$ML = KQ \text{ (opp sides equal KQLM)}$$

$$ML = HK \text{ (opp sides equal HKLM)}$$

$$\therefore HK = KQ$$

$$(ii) h = 48 - 41268995$$

$$= 48.4 \text{ m (3sf)}$$

$$(b)(i) x\text{-Int} \Rightarrow y=0$$

$$0 = 2x - x^2$$

$$0 = x(2-x)$$

$$x=0 \quad x=2$$

$$(ii) V = \pi \int_0^2 (2x - x^2)^2 dx$$

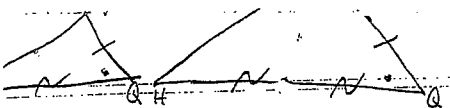
$$= \pi \int_0^2 (4x^2 - 4x^3 + x^4) dx$$

$$= \pi \left[\frac{4x^3}{3} - x^4 + \frac{x^5}{5} \right]_0^2$$

$$= \pi \left[\left(\frac{32}{3} - 16 + \frac{32}{5} \right) - (0) \right]$$

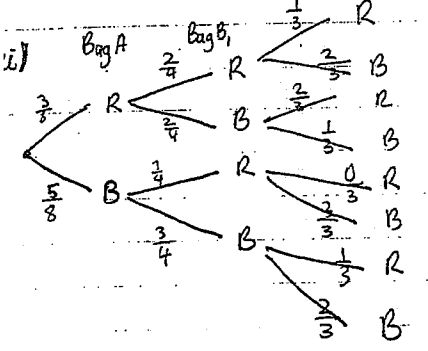
$$= \pi \left(\frac{16}{15} \right)$$

$$= \frac{16\pi}{15} u^3$$



$PQK = \angle LQH$ (common $\angle$)
 $\frac{QK}{PH} = \frac{QP}{QL} = \frac{1}{2}$ ($HK = KQ$)
 $\frac{LP}{PQ} = \frac{PQ}{QL}$ (side in ratio & included $\angle$ equal)
 $\Delta QPK \parallel \Delta QLH$ (corr. $\angle$ s in $\Delta QPK \parallel \Delta QLH$)
 $\angle PKQ = \angle LHQ$ (corr. $\angle$ s in $\Delta QPK \parallel \Delta QLH$)

i) $P(B) = \frac{5}{8}$



(both Blue from Bag B)
 $P(RBB) + P(BBB)$
 $\frac{3}{8} \times \frac{2}{4} \times \frac{1}{3} + \frac{5}{8} \times \frac{3}{4} \times \frac{2}{3}$
 $\frac{3}{8}$

QUESTION 9

i) $P(\text{no fruit}) = 1 - 0.75 = 0.25$
 n seedlings no fruit
 $= (0.25)^n$
 ii) 99% certain
 $P(\text{at least one}) = 0.99$
 At least one = 1 - none

$(0.25)^n \leq 1 - 0.99$
 $n \ln 0.25 \leq \ln 0.01$
 $n \geq \frac{\ln 0.01}{\ln 0.25}$

$n \geq 3.32$
 $\therefore n = 4$

(b) i) (a) $BC = r\theta$
 (b) $A = \frac{1}{2} r^2 \theta$

ii) (a) Given $P = 2r + r\theta = 12$
 $r(2 + \theta) = 12$
 $r = \frac{12}{(2 + \theta)}$

$Y = \frac{1}{2} \left(\frac{12}{2 + \theta} \right)^2 \theta$
 $= \frac{1}{2} \left(\frac{144}{(2 + \theta)^2} \right) \theta$
 $= \frac{72\theta}{(2 + \theta)^2}$

(b) Max $\Rightarrow \frac{dY}{d\theta} = 0$
 $Y = \frac{72\theta}{(2 + \theta)^2}$

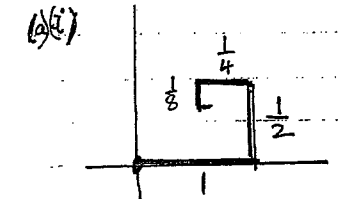
$\frac{dY}{d\theta} = \frac{(2 + \theta)^2 \cdot 72 - 72\theta \cdot 2(2 + \theta)}{((2 + \theta)^2)^2}$
 $= \frac{72(2 + \theta)^2 - 144\theta(2 + \theta)}{(2 + \theta)^4}$
 $= \frac{72(2 + \theta)[(2 + \theta) - 2\theta]}{(2 + \theta)^4}$
 $= \frac{72(2 - \theta)}{(2 + \theta)^3}$
 $\therefore 0 = \frac{72(2 - \theta)}{(2 + \theta)^3} \therefore \theta = 2$

where $\theta = 2$

θ	1	2	3
$\frac{dy}{d\theta}$	$\frac{72(2-1)}{(2+1)^3}$	0	$\frac{72(2-3)}{(2+3)^3}$
	$\frac{72}{3} > 0$	0	$\frac{-72}{125} < 0$

$\therefore$ Max value at $\theta = 2$
 At $x = 2$ $Y = \frac{72(2)}{(2+2)^2}$
 $= \frac{144}{16}$
 $= 9m^2$

QUESTION 10



$\rightarrow \Delta P_1: 1, -\frac{1}{4}, \frac{1}{16}, \dots$ $a = 1, r = \frac{1}{4}$
 $\Delta P_2: \frac{1}{2}, -\frac{1}{8}, \frac{1}{32}, \dots$ $a = \frac{1}{2}, r = -\frac{1}{4}$
 $S_{1\infty} = \frac{1}{1 - -\frac{1}{4}} = \frac{4}{5}$
 $S_{2\infty} = \frac{\frac{1}{2}}{1 - -\frac{1}{4}} = \frac{2}{5}$
 $\therefore \left(\frac{4}{5}, \frac{2}{5} \right)$

ii) Ant will only approach point $\left(\frac{4}{5}, \frac{2}{5} \right)$ as both r_1 & r_2 ($-1 \leq r \leq 1$) imply a limiting sum.

$12 = 240$
 $M = 2000$
 (i) $A_{120} = 25000(1.006)^{120} - 2000(1.006^{120} - 1)$
 $= 162498.4953$
 $= \$162498.50$ (2dp)

(ii) $A_{240} = 250000(1.006)^{240} - 2000(1.006^{240} - 1)$
 $= -16881.16943$
 Since A_{240} is negative, then Robert has paid more than he owed, ie loan is in credit after 20 yrs. The loan was paid off in less than 20 years.

(iii) $A_n = 0$
 $0 = 250000(1.006)^n - 2000(1.006^n - 1)$
 $= 250000(1.006)^n - 333333.3(1.006)^n + 333333.3$
 $= 333333.3 - 83333.3(1.006)^n$
 $83333.3(1.006)^n = 333333.3$
 $(1.006)^n = \frac{333333.3}{83333.3}$
 $1.006^n = 4$
 $n \ln 1.006 = \ln 4$
 $n = \frac{\ln 4}{\ln 1.006}$
 $= 231.7415...$
 $= 232$ months

$\therefore 240 - 232 = 8$ months
 Loan was paid off 8 months early.